Continual Learning for World Models

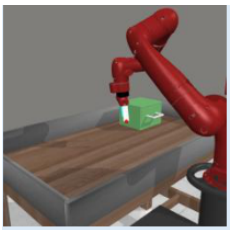
Reusable knowledge and two research routes

Research direction discussion

November 2026

One benchmark, three related tasks

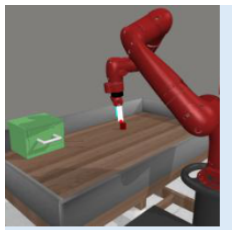
DrawerPnP: learn the primitives, then their composition



1. Pick and place a cube



2. Open a drawer



3. Open the drawer, place the cube inside

Reuse earlier interactions while keeping earlier abilities.

[1] Benchmarking World Models for Continual Learning on Compositional Tasks, 2026 preprint. Task order: Table II. Scenes: Figure 1, full-composition row, cropped.

Task goals and reusable predictions

Task-specific goals

Put the cube on the table.

Open the drawer.

Put the cube inside the drawer.

The goal changes the reward, the plan and the chosen actions.

Reusable predictions

How the gripper moves the cube.

How pulling moves the drawer.

How contact changes object motion.

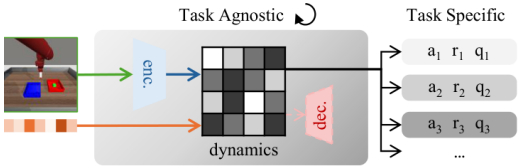
These relations can help with more than one goal.

Shared knowledge can support several goals.

[1] motivates task-agnostic prediction and task-specific control. The physical relations above explain the research aim; the benchmark does not prove the model discovers explicit physical rules.

The WM-CL goal: reuse without forgetting

A shared predictive model



The model predicts consequences.
Task heads use predictions for control.

In the drawer example

Keep: still place the cube after learning the drawer.

Reuse: use both earlier interactions when learning the combined task.

Grow: new experience improves knowledge useful for later goals.

Keep predictions useful as experience grows.

[1], Figure 2 crop. The paper evaluates forgetting and composition-task forward transfer. Knowledge growth and integration are the research aim, not conclusions established by one score.

Two routes to reusable world knowledge

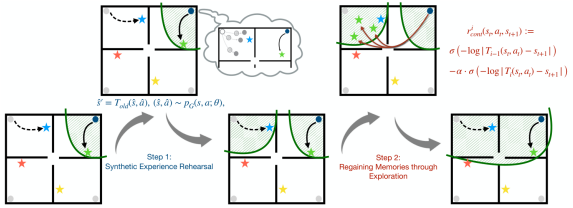
	Route A	Route B
Starting point	**Task-sequence experience**	**Pretrained visual or dynamics knowledge**
Main question	**Form and retain shared dynamics**	**Use and extend existing knowledge**
Main emphasis	**Sharing, memory and representation**	**Reuse and selective adaptation**

Architecture and pretraining can work together.

Conceptual research comparison. A learned architecture can later be pretrained; pretrained models can also need new learning mechanisms.

Route A: retaining learned dynamics

DRAGO: same dynamics, changing goals



1. Generate old state–action pairs.
Use old-model next-state predictions.

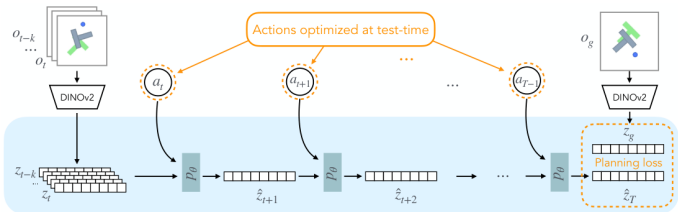
2. Revisit useful past regions.
Target gaps in current predictions.

New goals can expand coverage of the same shared world.

[2] Knowledge Retention in Continual Model-Based Reinforcement Learning, Figure 2 crop. Tasks change reward functions, not state space or dynamics.

Route B: reusing pretrained knowledge

DINO-WM ICML 2025



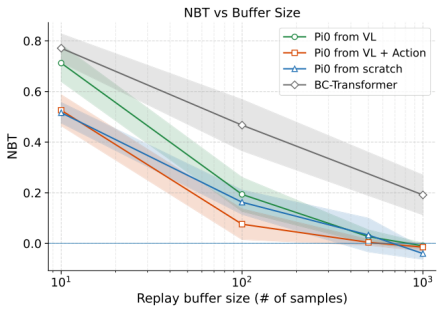
Frozen DINOv2 provides spatial visual features.

Learn how actions change those features.
Use a goal image to define a new target.

Reuse visual knowledge while learning action-conditioned dynamics.

[3] DINO-WM, Figure 2. Dynamics is learned from offline trajectories. Goal-based planning demonstrates this design, not continual-learning retention.

Pretraining can change continual learning



Less NBT means less forgetting.

VLA evidence ICML 2026 Oral

Same π_0 architecture, different pretraining.

With 1,000 replay samples per old task:
VL + action success: 86.3%
Scratch success: 65.5%

Low forgetting must be read together with task learning.

Pretraining and simple replay can support strong continual learning.

[4] Liu et al., Figure 4 and Table 4. Success values average three LIBERO suites at 1,000 transitions per old task. Pretrained models can still forget, especially with small replay.

Old performance can recover quickly

Updates to regain old peak

Percent of original training steps

LIBERO	π_0	BC-T
Spatial	6.6%	136%
10	10.5%	187%
Object	6.7%	180%
Goal	6.2%	33%

Recovery test

Relearn an old task after learning a later task.

Match data and optimization. Compare with first-time learning.

Full old-task data are available during recovery.

Performance loss need not mean all useful knowledge is gone.

[4] Liu et al., Figure 6(c), Figure 8 and Table 5. Ratios converted to percentages. Recovery probes retained task information; it does not prove unchanged representations or identify a unique failure mechanism.

Different routes, different bottlenecks

	Route A	Route B
Representation	Learn useful features across changing tasks	Check whether pretrained features cover the new physics
Main difficulty	Form shared dynamics while keeping old predictions	Extend existing knowledge without damaging useful predictions
Knowledge structure	Decide what to share, separate or rehearse	Decide what to freeze, condition or adapt
Research direction	Formation of reusable knowledge through experience	Growth of existing knowledge through adaptation

Research comparison by emphasis. Both routes face forgetting, knowledge access and integration; these are not exclusive assignments.

What each route offers my research

Route A: how knowledge forms

How do separate experiences form shared predictive structure?

Study learning mechanisms under controlled conditions.

Challenge: perception and control can mask the knowledge issue.

Route B: how knowledge grows

How can rich prior knowledge absorb new experience?

Study retention and reuse alongside genuinely new learning.

Challenge: hidden pretraining data can mask what is truly new.

Both routes serve my interest in reusable dynamics.

Presenter's research judgment. Choosing a pretrained starting point does not establish an algorithmic gap or make architecture research obsolete.

References

[1] H. Zhou, J. Watson, A. Lei, I. Posner.
Benchmarking World Models for Continual Learning on Compositional Tasks.
2026 preprint. Figures 1, 2; Table II.

[2] H. Fu, Y. Sun, M. Littman, G. Konidaris.
Knowledge Retention in Continual Model-Based Reinforcement Learning.
ICML 2025. DRAGO, Figure 2.

[3] G. Zhou, H. Pan, Y. LeCun, L. Pinto.
DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning.
ICML 2025. Figure 2.

[4] H. Liu, C. Kim, B. Liu, M. Liu, Y. Zhu.
Pretrained Vision-Language-Action Models are Surprisingly Resistant
to Forgetting in Continual Learning. ICML 2026 Oral.
Slide 8: Figure 4, Table 4. Slide 9: Figures 6(c), 8; Table 5.